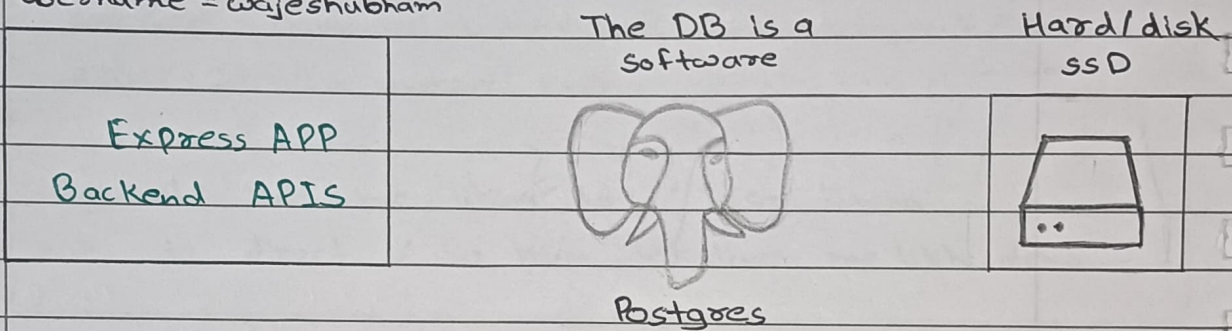


Chai aur Sql

1) Why we need database

- A database does not store a data within itself.
- for storing and managing data

query $\Rightarrow$ Get me a user jika
username = wajeshubham



- Software is something which is virtual
- so software does not store data
- Technically actual data goes to Hard disk / SSD

So, database separate from hard disk

When you say what is SQL or NoSQL you are actually talking about software

- under the hood both are storing data in same SSD.
- user's actual data in Hard disk

The query is

Give me a user with username wajeshubham

- So, there is two way's to get user data.

- first, removes the DB
and write a low level program to get the data from disk, means you run through the memory to find out where this field is located and where its value is and then get the data

But it's so complex when you have billions of rows means a lot of data.
So, it's not practical.

- Second, with DB

What DB does,

you give the query it internally creates a query plan, to find this data in memory but technically in disk,

- memory is ram

- disk is your persistent storage

So, it's plan the query, it's actually executes the query but under the hood, it's goes to disk and pick-up the data then return to the application

So, Why we used Database.

- to decouple your backend application with the actual hardware infrastructure

- basically it's called I/O input output

- When ever you want to add something to your hard disk or you want to get something from your hard disk.

That operation is very tough to do with programming language.

So, DB is dedicatedly handle all this things that's all database do.

Point to remember

- 1) Database doesn't actually store the data itself. Data get stored on hard disk.
- 2) DB is only for get and write the data.
- 3) all the database work this way. NoSQL and SQL under the hood you can't change the infrastructure 90% of time you have hard disk and SSD.

So, hard disk doesn't care who is database.

hard disk only have data which present in 1's and 0's

hard disk says,

Just tell me which locations you need data for and how much.

ex,

muje chahiye ek row jiska starting hai 0 (zero) and last hai 10

to me hard disk ko bol sakti hu ki hard disk me 0 se lekar 10 tak sara data lake do

- The lowest we can go in computer science is 2 0's
- and your hard disk actually ~~data~~ deals with it.

- It's doesn't know what is JavaScript, SQL & database
- ye sirf meje jo query bhejna hai, usko database se convert karke hard disk se data leke then application ko return karata hai.

SQL vs NoSQL

- Database who can understand SQL language, are SQL dbs
 - Databases who do not entertain SQL language are NoSQL dbs
 - Most of the SQL dbs (relational dbs) store data in table format
- RDBMS (Relational Database Management System)
ex Excel

Col1	Col2	Col3
row1 cell1	row1 cell2	row1 cell3
row2 cell1	row2 cell2	row2 cell3

- Jab bhi RDBMS dikhe, excel ko imagine karo. excel ke pass bhi columns hote hai, excel ke under bhi actual filter the columns, you can sort the column, you have datatypes in columns it's strict
- if you have number column you can only assign a number
- So, there is structured way to defining a data, you don't have flexibility.

Date: / /

What are NoSQL

ex,

- MongoDB

Let see as analogy

- Typescript / Javascript

- Typescript is strictly type language and Javascript is dynamically type language

So, simply

Typescript $\Rightarrow$ SQL

Javascript $\Rightarrow$ NoSQL

SQL is strict, if you give are giving it an initial definition,

you can not actually violate that definition

ex,

you have object in typescript

```
Const obj: { a: string; b: number; c: number }
```

```
Obj.c = 2
```

Note:- consider this is a sql table, app ne use bol diya ki a: string, b: number you can not add c column unless you add the definition c: number. This is sql database most likely Relational database, this is designed for durability and consistency

ACID Principle

Which actually ~~takes~~ talks about different aspects that relational databases cover

So, you can get durability and consistency, and you can actually do atomic transaction.

- Atomicity
- Consistency
- Isolation
- Durability

this makes your database rowbased.

that's why in banking sector or in critical sector ~~RDBMS~~ they use RDBMS mostly.

NoSQL

ex. In JavaScript

```
const Obj = {}
```

```
Obj = 2
```

Note:-

When you assign a mongoose schema to user, technically you can add new field without adding into mongoose schema.

How do you decide SQL or NoSQL

It's depends on the design decisions how is your application

ex.

you have chat message means you are sending a message. So, you want highest kind of speed or you don't want any strict structure.

If I want to add extra field, I want to be very fast.

that's mean I don't want to see any migration.

Then you can go for NoSQL.

But, If you have critical data, which you are dealing with,

so, you want durability and ACID compliance, this make sure,

your data never going to be corrupt.

this is specifically the selling point for your SQL database.

SQL USP: ACID compliance

most of the SQL not all are ACID compliance
ex, postgres is ACID compliance

In simple words,

Difference between SQL & NoSQL

SQL $\Rightarrow$ strict

NoSQL $\Rightarrow$ kind of dynamic

Note:-

What is migration?

- migration is something which you deal with, you are building application with nodejs or postgres, so, how does this

you define a table using some ORM like mongoose, sequelize ke bhi kuchh ORM hote hai, like Drizzle, pg-client & prisma

What do they do?

you can write javascript to define a table, but Database doesn't know the Javascript.

So, What Drizzle do.

Drizzle takes, Javascript,

converts it into SQL and then applies it to the database

that conversion and appliance that are migration. In simple words,

you are migrating your javascript changes into the SQL database.

ORM is Object Relational Mapping

ex, Drizzle is ORM

mongoose also ORM but actually ODM

ODM - Object Document Mapper

- a library that translate between the objects in your application code and the document stored in the database.

What is ORM

Let say,

you have Express and you define table in JavaScript object.

Then you have `col1: string`, `col2: number`,

So, what's ORM do

It takes this and converts it into a SQL code
then it's create a table then it's apply this
query to database

Express			
let obh = {	→ SQL code	Apply on	
col1: string,	CREATE TABLE	DB	
col2: number	obh (...);		
}			

- aapne likha JavaScript But ORM ne SQL me convert karke database me table bana diya.

What is SQL

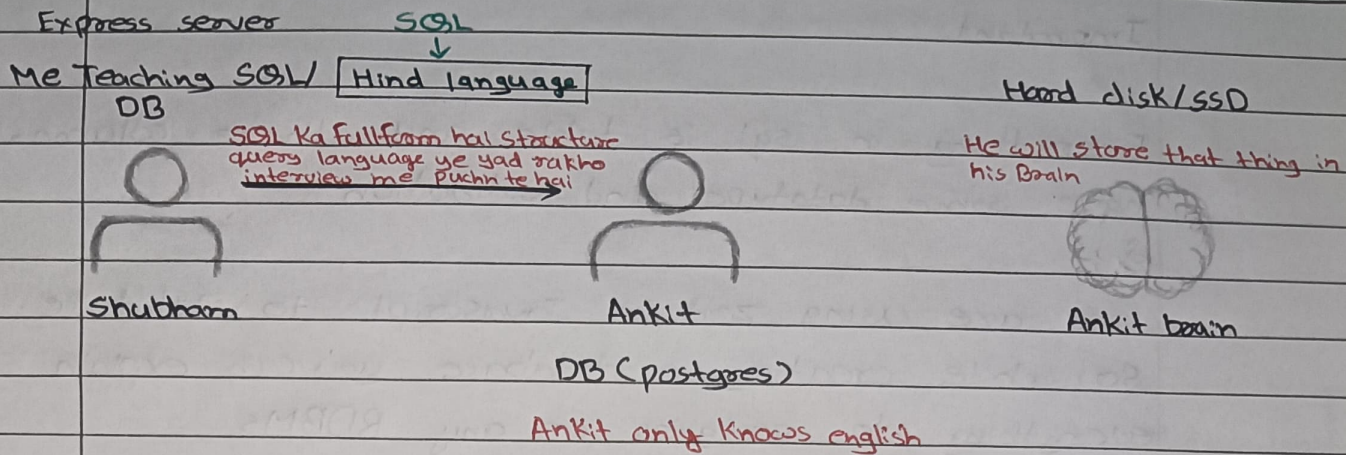
SQL ⇒ Structured Query Language

- This is a language, which database can understand.

Analogy:-

- Shubham teaching SQL in Hindi language,
Shubham say Ankit SQL ka fullform hai structure
query language ye yad rakho interview me puchh te hai

Date: / /



- fir Ankit usko sunega or apne brain me save karega
- Shubham ek express ka server hai
- Ankit Database hai (postgres)
- Hindi language SQL hai.
- par Ankit ko sirf yaha English aati hai to kya jo Shubham ki bat samaj paye ge.
- So, some database understand the language ~~him~~ ex here is hindi and this language is nothin but SQL
- This is just a high level language what you write which is sends to the database.
- Then db understands the context and performs the operations on your hard disk

Information

chat application are mostly based on NoSQL because it's dynamic, flexible

ex. discord is hav

discord is heavily rely on NoSQL and technically is fast too

Note: Views &

Materialized views
are advanced part

(It's like a writing in English language)

Date: / /

Important

Everything in sql is table and those columns have some datatype and validation.

ex,

we are using zod in Javascript to validate.

So, while creating a database you can explain everything to postgres or any RDBMS

Such as the structure of the database table and its validations.

make ensures that any data you insert in the future should adheres to these validations.

that's why its structured database.

01-Class-Sql

//To Create a table

```
CREATE TABLE students (
```

- This is a syntax for creating a table, this are reserve keyword in sql.
- Students are table name

Note:-

- If you write Students SQL automatically converts its to a students lowercase
- So when you use capitalise words make it in sure its in " " double quote.
- In SQL we mostly use snake case.
- you can't use " " single quote for naming its through error

Kebab case

ex, test-file-name.js

Note!- you can comment is sql
by using -- dash dash
ex, --name

Date: / /

Student-id SERIAL PRIMARY KEY,

↓
column name

- SERIAL is datatype, SERIAL is nothing but a integer means number and it's auto increment
- So, postgres automatically add 1, jese hi aap row insert karte ho.

example

- row insert $\Rightarrow$ Student-id 1
- row insert $\Rightarrow$ Student-id 2
- row insert $\Rightarrow$ Student-id 3
- When you insert a row, postgres automatically adds 1, 2, 3, ...
- For SERIAL starting value is one

- PRIMARY KEY is called as constraints and it's unique.
- It's get used when you join a table
- and it's never be empty

Note!-

- Serial auto increment integer, Primary Key is unique + never be null

first-name VARCHAR(50) NOT NULL

- VARCHAR is string
- 50 is the bite you are allocating
- that's mean you can only store up to 50 bite data
- if you don't define the size it's cover all the space VARCHAR

Note!-

- my first name should have max 50 letters

Note:- SERIAL is nothing but Int, it's just tell the Postgres it's int but isko autoincrement karu.

Date: / /

last-name VARCHAR(50),

- by default it is null, -- 50 * 8 bits
- For database by default any column who doesn't have null is null, empty kar sakte hai use.

email VARCHAR(322) UNIQUE NOT NULL,

- it's 322 because email might be more than 50 characters
- you can't add email twice when it's unique and not null
- email empty nahi ho sakta wo required field hai.

phone-number VARCHAR(10) UNIQUE NOT NULL,

- data type of phone-number never be a number
- phone number is 10 character per entry
- phone-number it's just a string there is no need to perform any operations, it's only for display.

age INT CHECK (age > 12),

- INT is the datatype who represents the number
- CHECK is for verification, ye column level par validation dal deta hai, ki jab bhi increment ho check us value ko leke comparison karega

country-code VARCHAR(4),

- maximum length of country code is 4.

Current-status VARCHAR(20) DEFAULT 'active'
CHECK (current-status IN ('active', 'graduated',
'dropped-out')),

- check current-status of students
- agar koi value na hai to default active hoga
- it's database validation.

Note:- you write array in sql on round bracket ()

- you don't have enum in every database,

- postgres have enum

- mysql have enum. But majority of database doesn't have enum

Date:

masterji_handle VARCHAR (50) UNIQUE,

-- masterji_username

has-joined-masterji BOOLEAN DEFAULT FALSE,

- Boolean either true or false

- But default is false

Current_Score INT DEFAULT 0 CHECK (Current_Score

≥ 0 AND Current_Score ≤ 100),

- If you add value of Score less than 0 or more than 100, it's going to throw error

enrollment-date DATE DEFAULT CURRENT_DATE

- It's a date field.

- CURRENT-DATE is a reserved variable in SQL

- When you insert now it's automatically fills the current date

-- This is UTC ('2026-04-04')); semi-colon to terminate the operation - end of query

// I want to add 8th column in same table

Syntax

ALTER TABLE students

operation → ADD COLUMN batch_name VARCHAR (50)

DEFAULT 'Web Dev 2020'

- ALTER means either remove or add column in table

Interview question

What is DDL?

- it's a Data Definition language
- it is language we used, it is the structured we used to define a structure in SQL database

ex,

CREATE $\Rightarrow$ defining a structure

ALTER $\Rightarrow$ changing or defining a structure again

- So, in simple words creating and altering table are DDL.

- you can use DDL as a part of SQL.

Q2 - class.sql

```
CREATE TABLE ipl-players (
    player-id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    team VARCHAR(50),
    role VARCHAR(50),
    runs-scored INT CHECK (runs-scored > 0),
    wickets-taken INT CHECK (wickets-taken > 0),
    auction-price-crores INT
);
```

```
ALTER TABLE ipl-players
```

```
ADD COLUMN nickname VARCHAR(50);
```


Date: / /

```
INSERT INTO ipl-players (name, team, role,  
runs_scored, wickets_taken,  
auction_price_crores) VALUES
```

```
('Virat Kohli', 'RCB', 'Batsman', 973, 1, 15.00, 'King Kohli'),  
( 'MS Dhoni', 'CSK', 'WicketKeeper', 450, 1, 12.00, 'Thala'),  
( 'Jasprit Bumrah', 'Mumbai Indians', 'Bowler', 15, 27, 12.00,  
  'Jassi'),
```

```
('Hardik Pandya', 'Mumbai Indians', 'All-Rounder', 400, 15, 15.00,  
  'Kung Fu Pandya'),
```

```
('Sunil Narine', 'KKR', 'All-Rounder', 350, 20, 8.50,
```

```
  'Carrom King'),
```

```
('Rohit Sharma', 'Mumbai Indians', 'Batsman', 550, 1, 16.00,  
  'Hitman');
```

```
('Rashid Khan', 'Gujarat Titans', 'Bowler', 50, 19, 15.00,  
  'The Magician'),
```

```
('Rinku Singh', 'KKR', 'Batsman', 475, 1, 0.55, 'The Spirit')
```

```
('Arjun Tendulkar', 'Mumbai Indians', 'Bowler', 10, 1, 0.30,  
  'Arjun'),
```

```
('Kane Williamson', 'LSG', 'Batsman', 600, 1, 11.00, 'Kane  
  Mama'),
```

```
('Mystery Player', NULL, 'Batsman', 1, 1, 1.00, 'Mystery man');
```

// How to run and select query

```
SELECT * FROM ipl-players;
```

- Select everything from ipl players

// you can select a specific column

```
SELECT name, nickname, team FROM ipl-players
```


--! Filtering

abhi team ke players chahiye jo is salt ipl jitenge

```
SELECT * FROM ipl-players WHERE team = 'Mumbai Indians';
```

ab mujhe use players chahiye jinka price 10 crores se jada ho, or unka name, nickname, price auction-price-crores

```
SELECT name, nickname, auction-price-crores FROM ipl-players WHERE auction-price-crores < 10;
```

you can also use other operators <, >, ≤, ≥, =

--! LOGICAL OPERATORS (AND, OR)

ab mujhe all-rounder chahiye jisne 10 se jada wicket liye hai

```
SELECT * FROM ipl-players WHERE wickets-taken > 10 AND role = 'All-Rounder';
```

ab mujhe yesi team chahiye jo is year hamre wali hai.

```
SELECT * FROM ipl-players WHERE team = 'Csk' OR team = 'RCB';
```


Date: / /

--! Pattern matching

muje ase players chahiye jinka name ka 3rd letter a ho (random requirement)

```
SELECT * FROM ipl-players WHERE name LIKE '--a%';
```

- this is used for analysis data

"Is this pattern in my rows"

un teams ko select karo jo layoff ne pahuche gi

```
SELECT * FROM ipl-players WHERE team IN  
( 'Mumbai Indians', 'KKR', 'Gujarat Titans', 'CSK', 'RCB' )
```

IN $\Rightarrow$ Include (ha; Ki nahi) aapka array

vo players jinka price 10-15 ke bich hai

```
SELECT * FROM ipl-players WHERE auction-price-  
crores BETWEEN 10 AND 15;
```

CSK ko chhodke sare pt. teams

```
SELECT * FROM ipl-players WHERE team  $\neq$  'CSK';
```

- alternate symbol for not $\diamond$ as $\neq$

--! Sorting

Which player has highest sold

```
SELECT name, nickname, auction-price-crores
FROM ipl-players
```

```
ORDER BY auction-price-crores DESC;
```

used that for sorting

- DESC $\Rightarrow$ Descending order

- ASC $\Rightarrow$ Ascending order

multicolumn sorting

- team ascending order me ho or auction-price-crores Descending

```
SELECT name, nickname, auction-price-crores
FROM ipl-players
ORDER BY team ASC, auction-price-crores DESC;
```

--! Pagination

```
SELECT name, nickname, auction-price-crores
FROM ipl-players
ORDER BY auction-price-crores DESC
LIMIT 3;
```

- LIMIT :- pehle auction price se sort karega then pehle 3 hi row fetch karega database.

- agar offset na ho to pehle 3 hi dega its assume ki offset zero hi hoga

Note 1-Offset is nothing but your skip

Date: / /

Offset 1- aap offset ko jo bhi number doge jese
Limit 3 to us utne rows skip kar dege

```
SELECT name, nickname, auction-price-crores  
FROM ipl-players  
ORDER By auction-price-crores DESC  
LIMIT 3 OFFSET 3;
```

- pehle 3 rows skip karke ~~age~~ age ke 3 rows
print kar dege

--! Modifying data in runtime

ye data aapke database me nahi hai, but you can
transform this data then give it to user

Convert crores into lakhs

```
SELECT name, nickname, auction-price-crores,  
(auction-price-crores * 100) AS price-in-lakhs  
FROM ipl-players;
```

- AS is a alias, it is a label

matlab is Column ko run time pe ek nam dedena

- this is how you transform your data in runtime
because you don't want to save this value

In next ipl session i want to add 2 crores

```
SELECT name, nickname, auction-price-crores,  
(auction-price-crores + 2) AS new-price  
FROM ipl-players;
```


Date: / /

--! How you can get distinct values

```
SELECT distinct role FROM ipl-players;
```

* Same columns ke unique value degh

Interview question

What is DQL

- It's Data Query Language and used to retrieve and fetch data from a database without modifying it
- In simple word you just read data from disk

03-class.sql

// insert, update and delete the data

ex.

let's say, chaicode me ek canteen khol di

```
CREATE TABLE canteen-menu (  
  item-id SERIAL PRIMARY KEY,  
  item-name VARCHAR(100),  
  category VARCHAR(50),  
  price int,  
  is-available BOOLEAN DEFAULT TRUE  
)
```

// let say how to insert data

Note:- There is no need to add custom id.
Primary key can handle that automatically.

Date: / /

INSERT INTO canteen-menu

(item_name, category, price) ← Column jisme value insert
karni hai.

VALUES ← insert value in that column

('Masala chai', 'Beverages', 10),

('Vada Pav', 'Snacks', 15),

('Samosa', 'Snacks', 12),

('Rajma chawal', 'Meals', 60),

('Maggi', 'Snacks', 25),

('Ice Tea', 'Beverages', 40);

('Dli', 'Snacks', 50);

SELECT * FROM canteen-menu;

// update the value of Vada pav

UPDATE canteen-menu

SET price = 20

WHERE item-name = 'Vada Pav';

SELECT * FROM canteen-menu;

- you can also change using id like item-id = 2

- this is called as mutation

// I want to reduced 5 in Beverages

UPDATE canteen-menu

SET price = price - 5

WHERE category = 'Beverages';

SELECT * FROM canteen-menu;

Note: DELETE & TRUNCATE are same
dono hi table khali hi karte hai
Deleting data doesn't affect the structure Date: / /

ab samosa khatam ho gaya hai, so is available
false karna hai and reduced its price too.

UPDATE canteen-menu

SET is-available = FALSE, price = 10

WHERE item-name = 'Samosa';

SELECT * FROM canteen-menu;

// Delete the idli

DELETE FROM canteen-menu

WHERE item-name = 'Idli';

SELECT * FROM canteen-menu

Interview question

DML - Data Manipulation language

- edit, add & delete row-level data from database
- Simply manipulate the data.

Dry run in sql

SELECT * ~~for~~ FROM canteen-menu

WHERE item-name = 'Samosa';

- is se aap verify karte hai ki where clause jo
lagi hai wo usi row pe hi lag rahi hai ki nahi
jab aap update karoge.
- hamesha har ek query pe dry run karke hi update or
delete operations karna

04- class .sql

// aggregation in sql

CREATE TABLE smart-watch-sales (

sale-id SERIAL PRIMARY KEY,

brand VARCHAR(50),

model VARCHAR(100),

city VARCHAR(50),

units-sold INT,

price-per-unit DECIMAL(10,2),

sale-date DATE

);

INSERT INTO smart-watch-sales

(brand, model, city, units-sold, price-per-unit, sale-date)
VALUE

('Boat', 'Storm call', 'Mumbai', 10, 1500.00, '2023-10-01'),

('Boat', 'Storm call', 'Delhi', 15, 1500.00, '2023-10-02'),

('Noise', 'Colorfit', 'Bangalore', 20, 2000.00, '2023-10-01'),

('Noise', 'Colorfit', 'Mumbai', 5, 2000.00, '2023-10-03'),

('Apple', 'Watch series 9', 'Mumbai', 2, 45000.00, '2023-10-01'),

('Apple', 'Watch series 9', 'Bangalore', 8, 4500.00, '2023-10-02'),

('Samsung', 'Galaxy watch', 'Delhi', 3, 25000.00, '2023-10-01'),

('Boat', 'Xtend', 'Pune', 25, 1200.00, '2023-10-04'),

('Noise', 'Pro 4', 'Delhi', 12, 2500.00, '2023-10-05');

- Whenever you are working with data analytics. Kitne rows, kis type ka data there are basic queries.

DSL are good for analysing and querying data mostly used in app level

ex,

getUserById, getUserByUsername this are plain queries.

But there are special queries jo analysis me jada kam aati hai or unhi ko hi aggregation bolte hai

matlab hum kuchh group karte hai some aspect based on same aspect to get some value, ex,

How much did Apple sell or Noise

How much was sealed at each location

To is type ke jo queries hote hai hum do aggregate and we used some sql syntax to get those result at run time,

ye kuchh bhi data hum store nahi karte,

- data always saved at flat level

How many rows in table in single number

SELECT COUNT(*) as total-rows FROM smart-watch-sales;

Note:- Don't do this in production, it can block your database

Count is aggregation function

// Kitna revenue hua vo chahiye
unit sold into price of individual unit

```
SELECT SUM(units_sold * price-per-unit)
  AS total-revenue FROM smart-watch-sales;
```

// average price kya he units ki

```
SELECT AVG(price-per-unit) AS avg-price-unit
  FROM smart-watch-sales;
```

record imbalace hai to mujhe average price di do units ki
to all over the store

// subse chipest smart watch chahiye smart watch ke sales
me

```
SELECT MIN(price-per-units) AS cheapest FROM
  smart-watch-sales;
```

// subse costly smart watch chahiye smart watch ke
sales me

```
SELECT MAX(price-per-units) AS costly FROM
  smart-watch-sales;
```

--! Group by

let see which brands sold unit solds is more than
my other brands unit.
simple konsa brand ka jada bika


```

SELECT brand, SUM (units-sold) AS
      total-units-sold
FROM smart-watch-sales
GROUP BY brand
ORDER BY total-units-sold DESC;

```

- Select or grouped by me same kora change name nhi to error dege
- SQL says: you can use aggregate function, but the column you demanding are not using aggregate function.
- you have to grouped your aggregate function based on that column
- In here aggregation is brand level
- If you group by brand is merged that and ignored other column
- This is most practical example which used in application level
- When you do sort based on location or sort based on brand we actually do this

//jis city se hamara sabse jada revenue hai vo select karo

```

SELECT city, SUM (units-sold * price-per-unit)
      as city-revenue
FROM smart-watch-sales
GROUP BY city
ORDER BY city-revenue DESC;

```

- This is also valid business requirement

-- Multiple column grouping

let say, maje mumbai pe city level and brand level pe kitne sold huve vo chahiye.

```
SELECT city, brand, SUM (units-sold) AS units  
FROM smart-watch-sales  
GROUP BY city, brand;
```

- It's only merged city and brand then sum the units-sold.

-- using HAVING Keyword

- It used to filter the results of a GROUP BY query based on the result of an aggregate function

```
SELECT brand, COUNT(*)  
SELECT brand, COUNT (price-per-unit)  
As min-price  
FROM smart-watch-sales  
GROUP BY brand  
HAVING COUNT (price-per-unit) > 2000;
```